DO NOW — RECALL

Lesson 120 · Following Algorithms and Building Trace Ta

Name: _____ Class: _____ Date: _____

1. What drives decisions in spiral development?

2. Why can an array provide direct indexed access?

3. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

4. For the binary value 1010, what is its decimal value?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Identification, assessment and mitigation of risk.

2. Model answer

Its element locations can be calculated from base address and index.

3. Model answer

Sum 0 and carry-out 1.

4. Model answer

10.

By the end, you can…

• Find variables, data structures, conditions and inputs.

• Update a table only when an assignment or state change occurs.

• Compare produced output against constraints, edge cases and intention.

• Trace recursive calls, identify a base case and compare with iteration.

Vocabulary — know these terms

trace table — Record of algorithm state as steps execute.

iteration — Repeated execution of instructions.

dry run — Manual simulation of algorithm.

boundary test — Test at edge of valid input domain.

recursion — A method calls itself with a smaller or simpler version of the problem.

base case — Stopping condition that returns without another recursive call.

NEW LEARNING

Read, annotate and explain

Identify state

Find variables, data structures, conditions and inputs.

Trace each step

Update a table when an assignment or state change occurs. For a recursive function, record each call , its argument and its eventual return value; unlike a loop, calls wait on the call stack.

Check outcome

Compare output against constraints and edge cases. Recursion must reach a base case that returns without another recursive call, or it cannot terminate safely.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace total ← 0; FOR n = 1 TO 4: IF n MOD 2 = 0 THEN total ← total + n; NEXT n; OUTPUT total.

Model answer

At n=1 total 0; n=2 total 2; n=3 total 2; n=4 total 6; output 6. Record n and total at each loop step.

Check: link each claim to the scenario and use precise OCR terminology.

Spot the stopping condition

n == 0 is the base case : return 0 immediately. Otherwise SumTo(n - 1) invokes the same method with a smaller argument. Assume n is non-negative.

Trace calls before returns

SumTo(3) waits for SumTo(2) , then SumTo(1) , then SumTo(0) . The first three calls remain active until the base case returns.

Follow the returns in reverse order

SumTo(0)=0 ; then SumTo(1)=1 , SumTo(2)=3 , SumTo(3)=6 . An iterative alternative would use a loop and running total; recursion instead retains pending calls.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace count ← 0; FOR i = 1 TO 5: IF i MOD 2=1 THEN count←count+1; NEXT; OUTPUT count. [6]

Question 2

Explain why initialising max ← 0 can cause a fault when finding the maximum of negative values. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Trace recursive SumTo(4) using the method shown, name its base case and explain one difference from an iterative solution. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A trace table only needs the final value of each variable.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.